EE 210L: Digital Circuits and Logic Design Lab

Lab 6: Adders and Comparators

Name: _____ Date: _____

Partner(s): _____ Section: _____

Record logic levels, 0 or 1, and the measured voltage where a column asks for it. Build the 74LS283 wiring table before you place the chip: its pin order is not sequential.

Part 5.1: Half Adder

A	B	V at S (V)	S	V at C (V)	C	Expected S	Expected C
0	0						
0	1						
1	0						
1	1						

Part 5.2: Full Adder

Draw the full adder circuit you built. Label every gate and pin.

Figure 1: Full adder built from gates.

A	B	C_{in}	V at S (V)	S	V at C_{out} (V)	C_{out}	Expected S	Expected C_{out}
0	0	0						
0	0	1						
0	1	0						
0	1	1						
1	0	0						
1	0	1						
1	1	0						
1	1	1						

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Two rows have S = 1 but different C_{out}. Name them, and say why a half adder cannot tell them apart.

Part 5.3: Four-Bit Addition (74LS283)

Wiring table first. A1/B1/Σ1 are the least significant bit.

Bit	A pin	B pin	Σ pin
1, LSB			
2			
3			
4, MSB			

#	A (dec)	A (bin)	B (dec)	B (bin)	Σ4 Σ3 Σ2 Σ1	C4	Sum as decimal	4-bit result correct?
1	9	1001	6	0110				
2	9	1001	11	1011				
3								
4								

Choose rows 3 and 4 yourself. At least one of them should overflow.

How does C4 tell you that a 4-bit sum has overflowed?

Part 5.4: Adder/Subtractor

Draw how the XOR gates and the control line connect to the 74LS283.

Figure 2: Adder/subtractor built around the 74LS283.

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Control	Operation	A	B	Σ4 Σ3 Σ2 Σ1	C4	Result as decimal	Expected
0	A + B	7	3				
1	A − B	7	3				
1	A − B	12	5				
1	A − B						

What does C4 mean when the circuit is subtracting?

Part 5.5: Magnitude Comparator (74LS85)

Cascade inputs for standalone operation: pin 4 = _____ pin 2 = _____ pin 3 = _____

#	A (bin)	B (bin)	A>B out	A=B out	A<B out	Correct relation?
1						
2						
3						
4						
5						
6						

Include one pair with A > B, one with A < B, one with A = B, and one pair that differs only in the least significant bit.

Part 5.5.1: Cascade Inputs Set Incorrectly

With pin 3 tied to GND and A = B, the outputs read: A>B = _____ A=B = _____ A<B = _____

Explain why the comparator behaves this way, and what the cascade inputs are actually for.

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Discussion and Conclusions

Summarize what you built and what it confirmed. Address at least: why a half adder cannot be used for any bit but the least significant, what ripple carry costs you as the adder gets wider, and how one chip manages to both add and subtract.

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